Problem 1 ... fair sprint

3 point

At this year's Olympic Games in Paris, the men's 100 m *sprint was won by American Noa Lyles in* 9.784 s, *Jamaica's Kishane Thompson finished second by five thousandths of a second Lyles ran on the track number 7, Thompson on the track number 4. Sprint is started by th report of the starter's gun firing. Every runner has a speaker right behind their starting block o ensure they all hear the sound at the same time. However, if they started at the sound of eal gunshot fired from a pistol located near lane 1 at the start line, by how much time woul Thompson have won? The width of a lane is* 1.22 m.

Jarda thought that Forrest Gump had won

Problem 2 ... rail problem

3 point

For a more comfortable train ride, the continuous welded rail *technology, which consists o welding the rails and creating a uniform surface for the train to pass, is used. However, suc track must withstand thermal expansion in summer and winter. What temperature differenc an a steel rail withstand if its linear thermal expansion coefficient is* $1.63 \cdot 10^{-5}\,\mathrm{K}^{-1}$ *and th permitted stress of the rail is* 600 MPa*? Assume that the tensile Young's modulus for stee s* 195 GPa*.*

David took a train to the dorms

Problem 3 ... in the shadows

3 point

Consider a street lamp of height $H = 3.2\,\mathrm{m}$ *as a point source of light. A person of height* $h =$
$= 1.8\,\mathrm{m}$ *is walking in a straight line directly away from the lamp at a constant speed of* $v =$
$= 1.5\,\mathrm{m\cdot s^{-1}}$. *Calculate the acceleration of the top of their shadow.*

Marek lives in the shadows

Problem 4 ... run as fast as you can

3 point

A pilot of a Mitsubishi A6M Zero fighter aircraft with a power of $P = 940\,\mathrm{hp}$ *was flyin horizontally over an atoll belonging to the United States of America at speed* $v = 180\,\mathrm{km}$ *When the pilot spotted the runway with aircrafts attempting to take off, he pulled the triggers o both* $7.7\,\mathrm{mm}$ *fixed-mounted Type 97 aircraft machine guns with a cadence of* $c = 900\,\mathrm{min}^{-1}$ *an a muzzle velocity of* $u = 745\,\mathrm{m{\cdot}s}^{-1}$*. Each of the machine guns had one full belt of ammunitio with 500 rounds. If the pilot flew exactly over the runway that was* $l = 7\,800\,\mathrm{ft}$ *long, how man rounds did he hit it with?*

Dodo watched war movie

Problem 5 ... water resource engineering

3 point

Water should be conserved, which is why we collect rainwater into a barrel. Water flows int he barrel via a vertical gutter, which collects water from the roof. When the barrel gets ful vater is wastefully spilled into the surrounding garden. We would like to redirect the water flo utside the barrel when it gets full, instead of overfilling the barrel. Verča's dad is an enginee just like she will be one day) and invented a clever machine to solve this problem. On th urface floats a plastic cylinder with a base of area $S = 100\,\mathrm{cm}^2$ and density $\rho = 550\,\mathrm{kg{\cdot}m}^{-3}$ onnected via rigid stick to the gutter end. When the surface rises, it pushes the cylinder up eventually closing the gutter end.

What is the minimal height of the cylinder in order for the buoyancy to be strong enoug o close the gutter end while being heavy enough to let the end open as the surface lowers Assume that the force $F = 15\,\mathrm{N}$ is required in order to open or close the gutter end (for closin up, for opening down), and otherwise the system cylinder-gutter end does not move.

Verča watched her daddy at work

Problem 6 ... cooking up acid

4 point

What is the minimum mass of the sulfuric acid H_2SO_4 *that we need to mix with* 200 ml *of pur water to bring the mixture to the boiling point, which is* 98 °C *in the laboratory? The initia temperature of both liquids is* 25 °C*. When mixed* 70.5 kJ *of heat is released per mole of sulfuri acid added.*

Assume a specific heat capacity of the resulting solution of $0.75\,\mathrm{cal\cdot g^{-1}\cdot °C^{-1}}$ *independent o external conditions, and with molar masses of hydrogen* $1.01\,\mathrm{g\cdot mol^{-1}}$*, oxygen* $16.00\,\mathrm{g\cdot mol^{-1}}$ *an sulfur* $32.07\,\mathrm{g\cdot mol^{-1}}$*. Assume that the reaction happens quickly and under standard conditions Neglect the heat capacity of the container and any heat losses. Next consider a simplified case where the amount of heat released per mole of added acid remains constant until the mole rati of acid to water exceeds* 0.1*. At that point, the amount of heat produced by adding more aci will decrease significantly, and therefore assume that in this case, the mixture will not reac the boiling point. If this occurs, give the result as* 0 g*.*

Karel was wondering, why is the mixing so dangerous

Problem 7 ... planespotting

3 point

Martin is looking out of the window and sees a flying plane. He wonders, how fast the plan could be flying. In his mind, Martin counts that the plane was in his field of view between tw houses for 10 seconds (the aircraft is in the field of view if Martin can see at least half of th aircraft). According to the rule "1 thumb in the distance of an outstretched hand correspond to 2.5 angular degrees" he calculates that the plane has flown 7.5 of his "thumbs" during tha time. Martin doesn't know how far the plane was flying, but he makes an estimation, that i his field of view, it occupied about half of his thumb during the whole observation and that th plane could be a Boeing 737-800. Calculate, how fast the plane was flying, based on Martin' estimations.

Martin was looking out of Veronika's windou

Problem 8 ... formula car on the ceiling

4 point

To maximize the frictional force between the wheels of Formula 1 cars and the road surface, th
ar body is designed so that air resistance pushes the car to the ground. At what speed woul
he Formula 1 cars have to travel in order to drive heads down on the ceiling? For simplicity
onsider that the car is a right-angled triangle of length 5.1 m, *height* 1.0 m, *and width* 1.8 m
The vehicle's mass, including the driver, is 800 kg. *Let us further assume that the air particle*
re initially stationary and elastically bounce off the vehicle.

Jarda wanted to skip a traffic jam in a tunne

Problem 9 ... warming up a baking tray with pendulum

4 point

The induced currents in a conductor can be demonstrated by an experiment in which a meta ray is placed under a swinging pendulum consisting of a magnet attached to a string and th oscillations are quickly damped. By how many degrees does such a tray heat up? We us n aluminum tray of width and length $a = 12\,\mathrm{cm}$, thickness $d = 0.42\,\mathrm{mm}$ with specific hea capacity $c_{\mathrm{Al}} = 896\,\mathrm{J{\cdot}kg^{-1}{\cdot}K^{-1}}$. The pendulum suspension has a length $l = 84\,\mathrm{cm}$, a maximun deflection $\alpha = 33\,^\circ$ and the mass of the weight is $m = 120\,\mathrm{g}$. Consider that half of the energ of the pendulum is converted into heat, which is received by the metal tray, and the other ha s dissipated by other means. The heat is distributed evenly in the tray. Neglect the weight o he string and the dimensions of the magnet. The density of aluminum is $\rho_{\mathrm{Al}} = 2\,700\,\mathrm{kg{\cdot}m^{-3}}$

Karel saw a picture of a demonstrative experiment in a textbook

Problem 10 ... diproton bids farewell

4 point

Let us imagine a particle composed of two protons, i.e., a helium 2 nucleus. Such a parti
le is highly unstable and rapidly decays into two separate protons. What is the maximun
peed that these protons can acquire after a spontaneous decay in the center-of-momentun
rame? Consider that the rest mass of a diproton is 2.015 89 Da *and the mass of a single proto*
s 1.007 825 Da.

Karel wondered about a problem with helium 2

Problem 11 ... accelerated deceleration

4 point

Jarda filmed his route around the city on a camera in his car. At home, he then bega to examine a section of the footage where he was traveling at the maximum allowed spee of $50\,\mathrm{km{\cdot}h^{-1}}$*, and* $55\,\mathrm{m}$ *before the intersection, the traffic light flashed red, so he slowed dow with constant acceleration to stop at the intersection. How many times faster does he hav to play this recording to make it appear as if he was slowing down with a drastic acceleratio of* $10g$*?*

Jarda wanted to prove he was a superhero

Problem 12 ... a cracked bottle

4 point

A bottle of sparkling water slipped out of Eliška's hand and fell to the ground. When it hit th loor, a hole was made in the bottle, from which the drink began to spurt. Consider a mode ituation where water sprays out at an angle $\alpha = 70\,^\circ$ *(measured from the ground) to a heigh f* $h = 1.20\,\mathrm{m}$*. What was the initial difference between the pressure in the bottle and th mbient pressure?*

Eliška dropped a bottle of Kofola

Problem 13 ... big guitar bang

4 point

As Paťo walked through the door, he slammed his guitar against the doorjamb. Later on, whe
he checked the damage, he noticed that one of the strings was tuned exactly a semitone highe
than the original frequency. Since the string was already due for replacement before that, Pat
figured he could kill two birds with one stone.

He decided to tightly and firmly wind a new layer of wire onto the string in the shape o
a helix. How thick of a wire, made from the same material, does he need to buy to retune th
string to its original frequency? The problematic string had a thickness of 1.1 mm *at the start*
Assume equal temperament tuning and that the tension in the string remains unchanged durin

his manipulation. Consider the string both in its original state and after the wire has bee
wound, as a homogeneous circular right cylinder. Pato didn't want to change the strings

Problem 14 ... road on road

4 point

Torrential rains carried away cube-shaped paving stones with an edge of $10\,\mathrm{cm}$ *and a mas of* $2\,\mathrm{kg}$ *from the road. What must have been the minimum depth of the water flow needed t overcome the coefficient of friction* 0.5 *and thus move the cobblestones lying separately on th subgrade? The water was flowing at a speed of* $1.3\,\mathrm{m\cdot s^{-1}}$*. Assume drag coefficient of* 1.05*.*

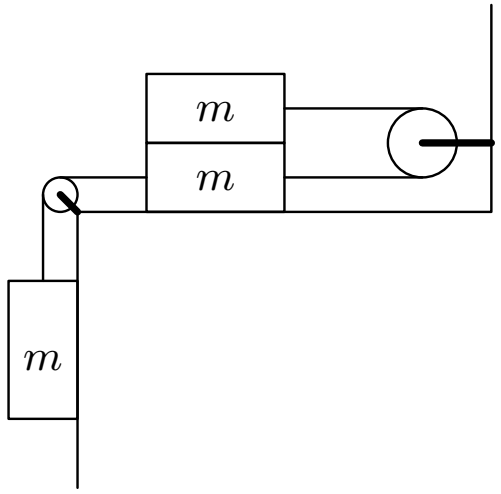
There were floods in Czechia this September

Problem 15 ... pulleys with blocks and friction

5 point

Two blocks lie on top of each other and are connected via a pulley so that the rope runs horizontally in both directions. The bottom block is connected by a rope via a pulley to another block which is hanging on this rope. There is a coefficient of friction $f = 0.10$ between the blocks and between the blocks and the ground. All three blocks weigh $m = 500\,\mathrm{g}$. What will be the magnitude of the acceleration of the hanging block? All the pulleys and ropes have no mass and the system is initially at rest.

Lego thought... this is definitely a physics problem.



Problem 16 ... increasing the degree of polarization

4 point

The degree of polarization of light P is defined using the intensities of two perpendicula polarizations, I_1 and I_2, as

$$P = \frac{|I_1 - I_2|}{I_1 + I_2}\,.$$

magine an environment in which the intensity of the vertically polarized light decrease o 99.0 % after each traveled distance L. Meanwhile, the intensity of the horizontally po arized light decreases faster, to 97.5 % after each distance L. After what distance will the ligh have a degree of polarization $P = 0.420$? Provide your answer as a multiple of L to one decima place. Before entering the environment, the light is unpolarized ($P_0 = 0.000$).

Karel was thinking about the anisotropic environment during the election

Problem 17 ... warmer background

5 point

Cosmic background radiation is one of the pieces of evidence for the Big Bang which we ca observe today. It corresponds very well to the thermal radiation of a black body with a tem perature of $T_0 = 2.725\,\mathrm{K}$. Given that the universe is $13.8 \cdot 10^9$ years old, how many years ag was the temperature of this radiation $T = 1.1T_0$? Assume that the universe expands uniforml (the scale factor is directly proportional to time).

Karel wondered if the universe is infinite... whether to have a toast...

Problem 18 ... combined

4 point

Consider two parallel infinite wires spaced $l = 10\,\mathrm{cm}$ *apart and non-conductive springs o*
stiffness $k = 10\,\mathrm{N{\cdot}mm^{-1}}$ *and a rest length* l *stretched between them. The distance betwee*
wo consecutive springs is $100\,\mathrm{km}$*. By how many millimetres does the spring elongate whe*
current of $20\,\mathrm{A}$ *starts to flow through one of the wires and* $10\,\mathrm{A}$ *through the other in th*
pposite direction? Assume that the system has reached a completely stabilized state.

Káťa wanted to come up with a problem that combined multiple branches of physics

Problem 19 ... laser lunch

5 point

n the third episode of the first season of the show The Big Bang Theory, Leslie Winkle heats he unch using a laser. In the episode, we learn, that it's a COIL laser ($\lambda = 1\,315\,\mathrm{nm}$) *with a powe of* $500\,\mathrm{kW}$ *and the heating process takes* $2.60\,\mathrm{s}$. *The heat capacity of her lunch is* $1.00\,\mathrm{kJ{\cdot}K^{-}}$ *nd it needs to be heated by* $60.0\,^{\circ}\mathrm{C}$, *to taste good. How many moles of photons emitted b he laser during the heating process do not contribute to the heating of her lunch? Assum hat if the photon transfers its energy to the lunch, there will be no energy losses.*

Terka was hungry in the lab

Problem 20 ... garage doors

4 point

A garage door weighs $m = 200\,\mathrm{kg}$. To keep the door motor from lifting the entire weight o the door, the door is connected to a steel torsion spring (see the image 1). The spring has a external diameter of $D = 4.1\,\mathrm{cm}$, and the steel wire which forms the spring has a diamete of $d = 5.0\,\mathrm{mm}$. One end of the spring is fixed. The other end is rotating together with th axis in which the cable is fixed to the bottom of the door. The door swings into the garage t horizontal position. Assume that the door is bending continuously. When the door is fully i the horizontal position, the spring is relaxed. The height of the door is $H = 2.1\,\mathrm{m}$. Determin the initial limiting number of windings N of the spring, for the door motor to exert utmost th force $F = 150\,\mathrm{N}$. For the given spring, the torque force M depends on the angle of rotation φ as

$$M = k\varphi = \frac{Ed^4}{64ND}\varphi\,,$$

where $E = 200\,\mathrm{GPa}$ is Young's modulus of elasticity for steel.

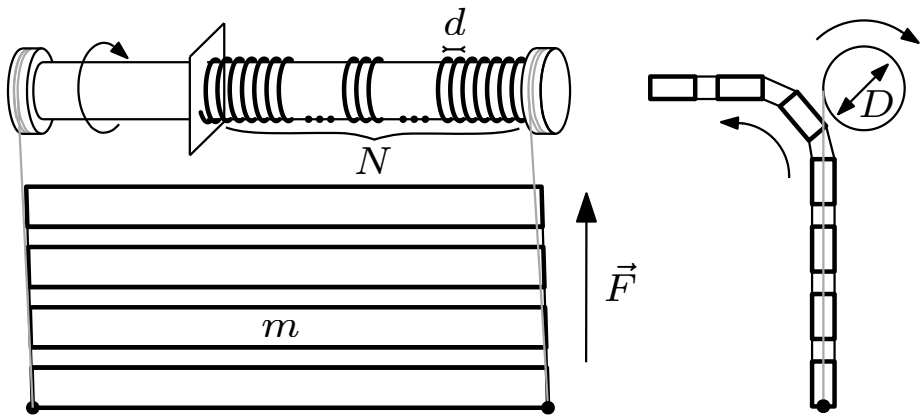


Figure 1: Scheme of the door mechanism.

Jarda couldn't figure out why the garage door wouldn't open when the spring broke

Problem 21 ... who would fear a buoy

5 point

A seagull weighing $M = 2.0\,\mathrm{kg}$ *lands on a cylindrical buoy of mass* $m = 5.0\,\mathrm{kg}$ *and base area* $S =$
$= 0.20\,\mathrm{m}^2$. *The not very graceful landing makes the buoy rock. Calculate the period of th*
mall oscillations of the buoy if it is oriented vertically, its top base is above the surface of wate
nd it is attached to the bottom of the lake by a taut chain through a spring of stiffness $k =$
$= 2.0\,\mathrm{kN{\cdot}m^{-1}}$. *The situation takes place on a lake.* *Vojta is not afraid of the buoy*

Problem 22 ... rolling glass of ice

5 point

Consider a tube (cylindrical surface) with very thin walls and a surface density of $\sigma = 5.5\,\mathrm{kg{\cdot}m^{-}}$
The tube is full of ice with density $\rho = 917\,\mathrm{kg{\cdot}m^{-3}}$*, the ice slides along the inner walls withou*
friction. We place the tube on a plane inclined at an angle $\alpha = 30\,^{\circ}$ *such that it can roll withou*
slipping, and its axis of rotation remains horizontal. What will be the acceleration of the tube

entre of gravity? The tube's radius is $r = 4.0\,\mathrm{cm}$ *and its length is* $l = 11\,\mathrm{cm}$.

Lego wondered about the acceleration of rolling

Problem 23 ... delicious fuel

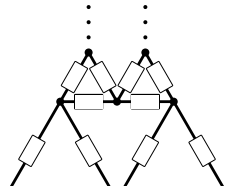
5 point

Mr. Simpson arrived to a czech nuclear power plant for a conference for nuclear safety engineers During a security check, the czech technicians found that Mr. Simpson had swallowed a nuclea reactor fuel pellet back at his home power plant. Let us assume that the pellet is made fro pure uranium oxide UO_2, which is enriched and is consists of 95 % of the isotope ^{238}U and 5 % o the isotope ^{235}U. What is the activity of uranium in the pellet, which Mr. Simpson swallowed The pellet has a cylindrical shape with a diameter $d = 7.5\,\mathrm{mm}$ and height $h = 11\,\mathrm{mm}$. Th density of uranium oxide is $\rho_{\mathrm{UO_2}} = 10.97\,\mathrm{g{\cdot}cm^{-3}}$. Jindra knows that one pellet is harmless

Problem 24 ... infinite resistance

5 point

Determine the resistance of an infinite network of ideal resistors between points A and B. Each resistor in the diagram has the same resistance $R = 1\,\mathrm{k}\Omega$. *Jindra likes triangles.*



Problem 25 ... live broadcast

5 point

We are recording a screen with a diagonal of $u = 23\,\text{inch}$ *and a resolution of* $1\,920\,\text{px} \times 1080\,\text{p}$
rom a distance of $d = 1\,\text{m}$ *using a camera. The camera feed is transmitted live back onto th*
creen. Assume that the camera captures a square area with a diagonal field of view $\alpha = 50\,^\circ$
hat the camera image is adjusted to fit the dimensions of the television (i.e., the originall
quare image is stretched to fill the entire screen), and that the camera is positioned along th
ymmetry axis of the screen. Determine how many times we will see the screen displayed withi
ts own image, given that the smallest discernible image of the screen must have an area of a
east $100\,\text{px}$ *to be distinguishable.*

Jarda got a new camera

Problem 26 ... periodic deflection

5 point

Electrons accelerated by a voltage of 1.0 kV *pass through the center of a deflection plate ca*
pacitor which is 3.0 cm *long. An alternating voltage of amplitude* 40 V *and frequency* 50 Hz *i*
connected to it. The electrons hit the screen of an oscilloscope located at 40 cm *from the en*
of the capacitor. What is the maximum speed of the beam's trace on the screen? The distanc
between the plates of the capacitor is 5.0 mm. *Neglect the electric field outside of it.*

Jarda admires modern flat screens

Problem 27 ... use crosswalks

5 point

Jarda decided to cross a road of width $5.0\,\mathrm{m}$ *when a car approached him in the middle of th*
ar lane at $50\,\mathrm{km\cdot h^{-1}}$. *The car was* $1.5\,\mathrm{m}$ *wide, driving in the middle of its lane, and had n*
ntention of braking. Jarda ran across the road in front of the car at a speed of $3.0\,\mathrm{m\cdot s^{-1}}$. *Hov*
lose to Jarda (along the road) could the car have been when he decided to step into the roa
f the car did not hit him? Jarda was running across the road in a straight line.

Jarda is not always using crosswalks. .

Problem 28 ... shot at a better future

5 point

From a cannon firmly attached to the ground, situated at the equator, we shoot a ball weigh ng 5 kg *at a* 45° *angle, which lands* 50 m *away from the cannon. How much would the siderea lay shorten if the Earth rotated with the same angular velocity as the ball immediately afte he shot? Assume the shot is fired westward, and approximate the Earth as a homogeneou phere. Consider only the gravitational effects of Earth.*

Monika is aiming to make fun of the problem-checkers

Problem 29 ... hanging phone

5 point

Petr's phone, weighing $m = 200\,\mathrm{g}$*, fell off a shelf while connected to a charger plugged int a wall socket. The charger cable consists of two copper wires with a circular cross-section, eac of length* $l_0 = 1.00\,\mathrm{m}$ *and diameter* $d_0 = 1.50\,\mathrm{mm}$*. The phone remained hanging on it afte the fall. By how many nanoohms did the resistance of the charger change? Assume that th charger forms a series circuit with the phone and the power source in the socket. The Young' modulus of elasticity for copper is* $E = 110\,\mathrm{GPa}$*, its Poisson's ratio is* $\nu = 0.340$ *, and its specifi electrical resistivity is* $\rho = 16.78\,\mathrm{n\Omega{\cdot}m}$*. Do not consider the effect of the plastic insulation o the cable.*

Petr's phone fell to the ground (again)

Problem 30 ... concentric spheres

5 point

Let there be an infinite number of concentric spherical shells with a uniformly distributed charge. The smallest sphere with a radius $r = 0.1\,\mathrm{mm}$ has a positive charge of $2Q$, where Q is the elementary charge. All the other spheres have a positive charge of Q, and the radius of the n-th sphere is n-times greater than the radius of the $(n-1)$-th sphere. What is the potential on the surface of the smallest sphere, assuming the potential at infinity is zero?

Monča found a potential use of some approximation

Problem 31 ... measured gas

5 point

Let us completly submerge a measuring cylinder into water, turn it upside down and pu
t out so that the part above the water's surface is filled with water. Then, we introduc
a tube beneath it, through which flows argon from a gas cylinder. The gas bubbles up int
he measuring cylinder, creating a gas pocket at the top. Once the argon stabilizes and th
water level inside and outside of the cylinder equilizes, we measure that the volume of the ga
s 0.85 l. *What is the mass of argon in the measuring cylinder? The temperature of both th*
water and the inside of the cylinder is 81 °C *and the external pressure is atmospheric.*

Jarda is again trying to catch you off guard this yea

Problem 32 ... weather watching

5 point

A meteorological satellite continuously captures images of the Earth's surface and transmits th data back to Earth at a constant frequency f. Jindra first picked up the signal from the satellit on his antenna when the satellite climbed just above the horizon. The frequency was $f_{\max}$ = = 137.915 51 MHz. The satellite subsequently climbed further above the horizon and reache its maximum altitude. The last signal Jindra received was just before the satellite went belo the horizon. The frequency at that time was $f_{\min} = 137.909\,49$ MHz. What is the radius of th satellite's orbit?

Assume that the satellite orbits the Earth in a circular orbit. Neglect the rotation of th Earth and its effect on the frequency shift. *Jindra was at the Expo*

Problem 33 ... overpriced taxicab

6 point

During the taxi ride, Monča realized it could not be profitable for any taxicab if the custome used its services for too long. However, she let her imagination run loose and fantasized abou a taxi that gradually increased its fare during the ride. Let us imagine a hypothetical tax operating as follows: for the first hour of the ride, it charges a fixed final price, and the charges this amount for the following 59 minutes of the ride. Then, the driving time, whic

osts this price, continues to decrease as a geometric sequence. Furthermore, this taxi travel
long segments of length s_0. The first segment takes 1 hour to traverse, the second one i
hour and 1 minute, and this travel time for the segment of length s_0 further increases a
geometric sequence.

Consider the shortest possible trip for which we would pay an infinite sum. Determine th
verall average speed of such a trip if the speed of the taxi is constant in each segment, wit
he taxi traveling at $30\,\mathrm{m\cdot s^{-1}}$ in the first segment. *Monča got scammed in a taxicab*

Problem 34 ... pressurized bubble in oil

5 point

What additional force would we need to apply on a piston, which leads into a tank containin
a small air bubble in oil, to reduce its radius to half? The piston has a cross-sectional area of $S =$
$= 4.20\,\mathrm{cm}^2$. *For simplicity, assume that the air in the bubble was initially at normal condition*
and its initial radius was $r = 0.420\,\mathrm{mm}$. *The surface tension of the oil is* $\sigma = 3.42 \cdot 10^{-4}\,\mathrm{N{\cdot}m^{-1}}$
The oil conducts heat well, and its temperature does not change significantly during the process

Karel was thinking about an undesirable bubble

Problem 35 ... dramatic escape

6 point

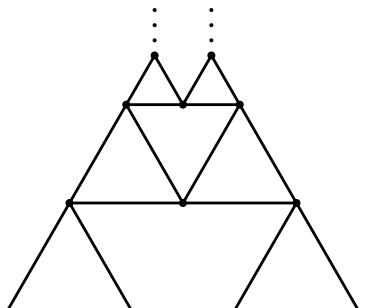
Marek is running away from his responsibilities across a bridge of height $H = 4.0\,\mathrm{m}$ *but realize he is surrounded. He decides to jump onto a boat of length* $L = 8.0\,\mathrm{m}$, *whose bow is jus emerging from under the bridge and is moving forward at a speed of* $v_\mathrm{l} = 5.0\,\mathrm{m{\cdot}s^{-1}}$. *What i the difference between the maximum and minimum angles* α *at which Marek can jump fro the bridge at a speed of* $v = 8.0\,\mathrm{m{\cdot}s^{-1}}$ *to land on the boat? Assume that Marek jumps upward and neglect the height of the boat.*

Marek is an outlaw. Jarda wants him to proofread this year's series

Problem 36 ... endless resistance II

6 point

Determine the resistance of an infinite network of wires between points A and B. The triangles forming the network are equilateral, and the distance between the points A and B is $a = 1.000\,000\,\mathrm{m}$. *All wires are of the same type and have a resistivity of* $\lambda = 1.000\,000\,\Omega\cdot\mathrm{m}^{-1}$. *Jindra is getting tangled up in the wires.*



Problem 37 ... dipole on a spring

5 point

Consider two charged particles which together form an electric dipole with a moment $p_0 =$ $= 2.20 \cdot 10^{-25}\,\mathrm{C \cdot m}$. *The system is in its energy minimum, where the particles are separate* *by a distance of* $l_0 = 3.10\,\mathrm{nm}$. *When displaced, they behave as they were connected by sprin* *of stiffness* $k = 4.80\,\mathrm{mN{\cdot}m^{-1}}$. *What would be the dipole moment if the dipole was moving a*

speed of $v = 28.0\,\mathrm{km{\cdot}s^{-1}}$ *perpendicular to the direction of its dipole moment in a homogeneou*
magnetic field with an induction of $B = 250\,\mathrm{mT}$*, which is perpendicular to both the dipole*
velocity vector and the direction of the dipole moment? Provide the highest possible value.

Jarda changes in the magnetic field

Problem 38 ... not-so-warm water

6 point

At a FYKOS trip, Petr stepped into the shower at midnight. When he turned on the water a fully filled boiler with a volume of $V = 100\,\mathrm{l}$ *started heating its contents. During this process water flows out to the shower at a rate of* $Q = 0.1\,\mathrm{l\cdot s^{-1}}$*, and cold water flows into the boiler at the same rate from the supply. The temperature of the cold water from the supply is* $T_\mathrm{s} = 20\,^\circ\mathrm{C}$ *which is also the initial temperature of the water in the boiler. How long will it take before Petr can shower with warm water at a temperature of* $T_\mathrm{k} = 40\,^\circ\mathrm{C}$*? The boiler has a power output of* $P = 15\,\mathrm{kW}$*. Assume that the incoming water mixes perfectly with the water in the boiler.*

Petr is not a cold water lover

Problem 39 ... hourglass

6 point

Let us consider an hourglass which, instead of sand, contains water. The apex angle is 40 °
and the gap between the chambers has a diameter of 2.0 mm. *What must be the volume of on*
chamber, if an initially fully filled chamber empties in 2 min *after being flipped? The entir*
double cone is rotationally symmetrical, and the two chambers are connected by a thin tub
along the side so that the pressure of the gases inside is not a factor. Neglect the surface tensio
of water and assume that the connection between the chambers is small compared to the othe
dimensions. *Jarda made use of Anika K's suggestion*

Problem 40 ... rectified efficiency

6 point

We have a Graetz (diode) bridge assembled with diodes. These diodes do not conduct an current until the voltage reaches $U_{\min} = 0.62\,\mathrm{V}$*, after which the current drops (and therefor he diode's resistance) becomes negligible. What is the maximum achievable efficiency (in % or a device connected to the rectified alternating current from this bridge, given that we us harmonic source with a maximum voltage of* $U_{\max} = 1.5\,\mathrm{V}$*? We want to compare this efficienc o the maximum efficiency of a circuit without any bridge. Karel was thinking about AC*

Problem 41 ... all the same

6 point

magine a planet where a sidereal day is as long as it is on Earth. Furthermore, on the latitud
vhere Prague is located on Earth, the gravitational acceleration is also the same. However, th
adius of the planet is ten times larger than the Earth's radius. What is its average density
Consider both planets to be perfectly homogeneous spheres.

Jarda was in a wrong classroom, but he didn't realize

Problem 42 ... suspended

6 point

Consider two suspension points at the same height at a distance $d = 50.0\,\mathrm{cm}$ *apart. On th*
first one, we hang a weight of mass $m_1 = 1.30\,\mathrm{kg}$ *from a suspension of length* $l_1 = 60.0\,\mathrm{cm}$. *O*
the other, we hang a weight of mass $m_2 = 2.10\,\mathrm{kg}$ *from a suspension of length* $l_2 = 45.0\,\mathrm{cm}$
We then connect both weights with a rope of length $l = 30.0\,\mathrm{cm}$. *At what distance below th*
suspension plane will the lighter weight be located? *Dodo was hanging the laundry*

Problem 43 ... border hill with a lookout tower

7 point

Lego is at a lookout tower between the Czech Republic and Slovakia. He hangs a pulley on th tower, over which he throws a rope, with carts at the ends having masses $m_{\mathrm{C}} = 24\,\mathrm{kg}$ (place on the Czech side) and $m_{\mathrm{S}} = 15\,\mathrm{kg}$ (on the Slovak side). The hill has a slope of $\alpha_{\mathrm{S}} = 10\,^{\circ}$ on th Slovak side and $\alpha_{\mathrm{C}} = 14\,^{\circ}$ on the Czech side. The rope forms angles $\beta_{\mathrm{S}} = 15\,^{\circ}$ and $\beta_{\mathrm{C}} = 11$ with the hill. What is the acceleration of the cart on the Slovak side, assuming all friction i neglected? If the cart moves upward toward the Czech side, enter a positive value; if downward a negative one. The pulley and rope are massless, and the rope is perfectly inelastic.

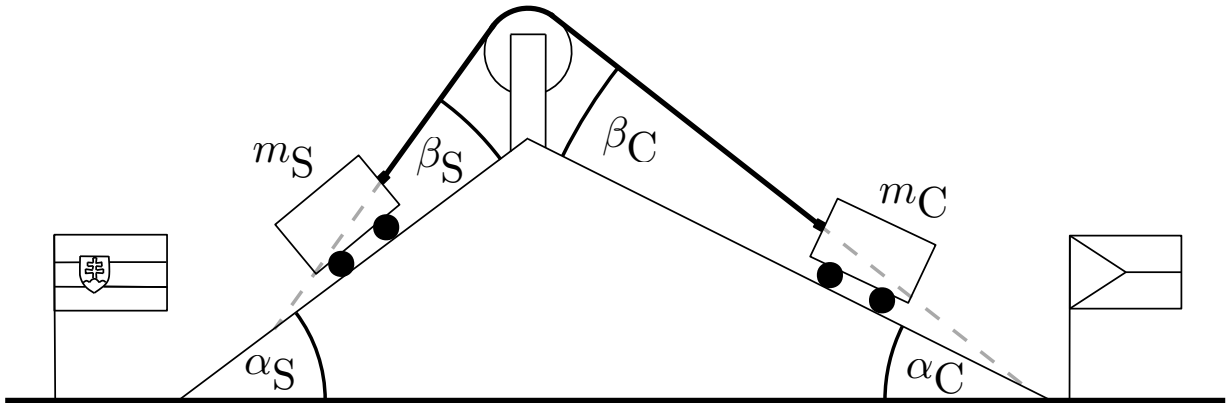


Figure 4: The scheme of the hill.

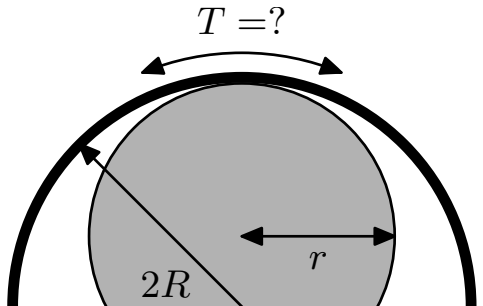
Lego, of course, imagined the vehicles from FYKOS camp

Problem 44 ... loose ring

6 point

A thin ring with radius $r = 5.0\,\mathrm{cm}$ is threaded onto a fixed horizontal rod with a circular cross-section of radius $R = 2.0\,\mathrm{cm}$. Determine the period of the small oscillations after a displacement from the equilibrium position. The ring does not slip relative to the rod.

Jarda is already thinking about a wedding.



Problem 45 ... double detection

7 point

Jindra is measuring the decay of a radioactive isotope using an energy-sensitive detector. H knows that one decay releases the energy E. During the measurement time $T = 300\,\mathrm{s}$ he detect $N_1 = 9\,728\,350$ events with energy E and $N_2 = 166\,545$ events with energy $2E$. The numbe of events with energy $3E$ or higher is negligible. Particle detectors have a so-called dead tim during which they process the signal from the previous particle. Specifically, Jindra's detecto also records particles that arrive during the dead time, but it cannot detect them as separat particles, instead, it adds their energy to the first particle. What is the dead time of Jindra' detector?

Jindra is counting decays

Problem 46 ... back to the U-tube

6 point

Let us consider a U-tube with an internal cross-sectional area $0.80\,\mathrm{cm}^2$*, into which we pour* 110 m
of water. The length of each of the symmetrically placed arms of the tube is 90 cm*. We plac*
an inflated balloon over one end of the tube, causing the water level in the other arm to ris
by $\Delta h = 4.0\,\mathrm{cm}$*. How much do we need to compress the balloon, i.e., by how much do we nee*
to decrease its volume, so that the water starts overflowing from the other arm? The volume o
the balloon just before compression is $V_0 = 750\,\mathrm{ml}$*. Neglect the volume of the horizontal par*
and the curved sections of the U-tube. *Jarda was devising an experiment for FYKOS*

Problem 47 ... acceleration of an idealized car

6 point

Let us consider a car with a power output of $P = 44\,\mathrm{kW}$ *and a mass of* $m = 1\,400\,\mathrm{kg}$*, whic*
s subjected to a total resistive force of $F = 950\,\mathrm{N}$ *(we assume this force is constant – as a*
ow speeds, air resistance is negligible). In how much time will the car accelerate from 0 *t*
$_{\mathrm{f}} = 18\,\mathrm{km{\cdot}h^{-1}}$*?*

Lego sometimes ponders like this

Problem 48 ... fishing

7 point

A point particle with a mass of $m = 1.85\,\mathrm{kg}$ *lies on a horizontal surface. It is connected to a tigh*
rope that ends at a winch located at a horizontal distance $L = 20.4\,\mathrm{m}$ *from the point particl*
and at a height $H = 6.35\,\mathrm{m}$. *The winch starts winding the rope at a speed of* $v = 3.25\,\mathrm{m{\cdot}s^{-1}}$
How long will it take for the point particle to lift off from the surface if the coefficient of frictio
between it and the surface is $f = 0.415$? *Jarda was winding up a cable from his mowe*

Problem 49 ... the oppressed air

8 point

What is the minimal volume at temperature $t_v = 25.0\,°C$ to which $V_1 = 1.00\,m^3$ of air a normal pressure p_a and at same temperature can be compressed? We have a water boiler c volume $V = 100\,l$ and temperature $t_b = 80.0\,°C$ as an energy source, and a large lake of wate at a temperature $t_j = 5.00\,°C$ as a cooler. Assume that the air behaves as a monatomic idea gas. All hypothetical machines used to compress the gas must return to their original state a the end (it is not necessary to consider the specific design of these auxiliary machines to solv the problem).

Dodo needed compressed air

Problem 50 ... optimally excited quantum oscillator

7 point

Lego created a one-dimensional quantum linear oscillator with $\omega = 1.17 \cdot 10^{13}\,\mathrm{s}^{-1}$. *He connect*
t to a thermal reservoir of temperature T *and waits until equilibrium is reached. What shoul*
his temperature be to maximize the probability of the electron occupating the first excite
tate? According to Lego, the name of this problem is very photosynthetic

Problem X.1 ... floating solar sail

4 point

What would the surface density of a solar sail need to be for the pressure of the solar radiatio
o exactly balance the gravitational force of the Sun? The surface of a solar sail is perfectl
eflective, and the plane of the sail is perpendicular to the Sun.

Jindra was interested, what area he would need to expand to freely float in the Solar system

Problem X.2 ... solar sail on the way to the stars

5 point

Consider a solar sail of mass $m = 10\,\mathrm{kg}$ *and a sail of area* $S = 10\,000\,\mathrm{m}^2$. *The sail begin ts journey in Earth's orbit with zero initial velocity relative to the Sun. What will be th magnitude of the speed of the solar sail at infinity? The surface of a solar sail is perfectl eflective, and the plane of the sail is always perpendicular to the Sun.*

Jindra wanted to expand so much, that he could fly to the stars

Problem X.3 ... solar sail on the way to the stars II

6 point

Consider a solar sail of a mass of $m = 10\,\mathrm{kg}$ *and a sail of area* $S = 10\,000\,\mathrm{m}^2$. *Initially, th olar sail orbits the Sun along Earth's orbit, but far enough to be outside Earth's gravitationa nfluence. Suddenly, it unfurls its sail very quickly. What will the speed of the solar sail be a nfinity? The surface of the solar sail is perfectly reflective, and the plane of the sail is alway riented perpendicular to the Sun. Assume Earth's orbit is circular.*

Jindra felt that the Earth was attracting him too strongly

Problem X.4 ... solar sail on the way to Mars

7 point

What is the necessary surface density of a solar sail, in order for it to travel from Earth's orbi o Mars' orbit via a Hohmann transfer orbit? The solar sail initially orbits the Sun in th ame orbit as Earth, then deploys its sail. The surface of the solar sail is perfectly reflective nd the sail plane is always oriented perpendicular to the Sun. The solar sail has no rocke ngines and relies solely on its reflectivity. Ignore Earth's gravitational influence at the star f the solar sail's journey, and assume the orbits of both Earth and Mars around the Sun ar ircular. Jindra wanted to surprise Teri with a problem about the Hohmann transfer orbi

Problem H.1 ... air under water – simplified to the maximum

4 point

We have a cube of side length $a = 10\,\mathrm{cm}$ *and density* $\rho_\mathrm{k} = 150\,\mathrm{kg{\cdot}m^{-3}}$. *Starting from a positio where the bottom face of the cube is just touching the water surface, we submerge the cube i the water by pushing it straight down (without any rotation) by* $h = 6.0\,\mathrm{cm}$. *How much work i performed in this process? Assume that the surface area* S *of the water is enormous* ($S \gg a^2$) *so the water level does not change during the process.* *Lego was correcting..*

Problem H.2 ... air under water – very simplified

4 point

We have a cube of side length $a = 10\,\mathrm{cm}$ *and density* $\rho_\mathrm{k} = 150\,\mathrm{kg{\cdot}m^{-3}}$. *Starting from a positio where the bottom face of the cube is just touching the water surface, we submerge the cube i the water by pushing it straight down (without any rotation) by* $h = 15\,\mathrm{cm}$. *How much work i performed in this process? Assume that the surface area* S *of the water is enormous* ($S \gg a^2$) *so the water level does not change during the process.* *...and as Lego was correcting..*

Problem H.3 ... air under water – less simplified

5 point

We have a cube of side length $a = 10\,\mathrm{cm}$ *and density* $\rho_\mathrm{k} = 150\,\mathrm{kg{\cdot}m^{-3}}$. *Starting from a positio*
where the bottom face of the cube is just touching the water surface, we submerge the cube i
water container by pushing it straight down (without any rotation) by $h = 15\,\mathrm{cm}$. *How muc*
work is performed in this process? The cross-section of the container (i.e. the surface area
s $S = 300\,\mathrm{cm}^2$.

...Lego thought of different simplified versions of the problem..

Problem H.4 ... air under water – still simplified

5 point

We have a cube of side length $a = 10\,\mathrm{cm}$ *and density* $\rho_\mathrm{k} = 150\,\mathrm{kg{\cdot}m^{-3}}$*. Starting from a positio*
where the bottom face of the cube is just touching the water surface, we submerge the cub
n a water container by pushing it straight down (without any rotation) by $h = 6.0\,\mathrm{cm}$*. Hov*
much work is performed in this process? The cross-section of the container (i.e. the surfac
rea) is $S = 300\,\mathrm{cm}^2$*.*

...until Lego finished correcting

Problem M.1 ... first ride through Blanka

3 point

A road pirate drives a car at $v_1 = 80.0\,\mathrm{km{\cdot}h^{-1}}$. *After travelling one third of the distance ove which the section measurement is taking place, he realises that he should maintain the maximu permissible speed* $\bar{v} = 70.0\,\mathrm{km{\cdot}h^{-1}}$, *which is measured as the average speed of travelling throug the measured section. What is the highest speed* v_2 *he can travel for the next two-thirds of th rack to maintain the maximum permissible speed? Neglect the time required to decelerate t he desired speed. Give the answer in* $\mathrm{km{\cdot}h^{-1}}$.

Karel was thinking about speed

Problem M.2 ... riding through Blanka... maybe... 4 point

magine that you need to decide how to travel to your cottage. You are choosing betwee route of length $s_1 = 13\,\mathrm{km}$ *traveled in* $t_1 = 25\,\mathrm{min}$ *and a route of length* $s_2 = 20\,\mathrm{km}$ *travele* $t_2 = 20\,\mathrm{min}$. *Assume that the gasoline consumption is constant* $c = 6.7\,\mathrm{l}/(100\,\mathrm{km})$ *and th rice of gasoline is* $p = 40.2\,\mathrm{CZK}\cdot\mathrm{l}^{-1}$. *It is said that time is money. What is the minimum valu our time must have to make it more profitable for you to take the longer route? Give th esult in Czech crowns per hour. Karel was thinking about finances and the value of time*

Problem M.3 ... changing speeds in Blanka

4 point

magine you are driving on a straight road with no restrictions due to traffic, but you want t dhere to the speed limits. On the road, the speed limits regularly alternates between v_{50} = = $50\,\mathrm{km\cdot h^{-1}}$ *and* $v_{70} = 70\,\mathrm{km\cdot h^{-1}}$ *and the signs are always* $d = 500\,\mathrm{m}$ *apart. You drive i uch way, that at the sign 70, you start to accelerate with acceleration* $a = 1.2\,\mathrm{m\cdot s^{-2}}$ *from th peed* v_{50} *to* v_{70}*. Conversely, you start decelerating with an acceleration of the same magnitud ut in the opposite direction to reach a speed of* v_{50} *just at the 50 sign. Otherwise, you alway drive at the maximum speed allowed in the section. What average speed in kilometers per hou will you reach? Ignore the length of the car.*

Karel thought about average

Problem M.4 ... riding through Blanka... pirate style

5 point

magine a road pirate, with a car of mass $m = 1\,290\,\mathrm{kg}$ and of engine power $P = 92.0\,\mathrm{kW}$, whic drives at speed $v_0 = 70.0\,\mathrm{km{\cdot}h^{-1}}$ into the Blanka tunnel complex and decides to accelerate wit maximum engine power. What speed would he achieve if he could accelerate this way for th ntire length of the complex $d = 5\,502\,\mathrm{m}$? For simplicity, neglect the tunnel height changes an he drag forces (which in reality play an important role) and assume that the car has no limi n its maximum speed. Give the result in kilometers per hour.

Karel thought about car acceleration